

EXPERIMENT NO. 1

STUDY OF MICROWAVE COMPONENTS & EQUIPMENT

Date: 19/8/2025	Group No. 2
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Aim: To become familiar with microwave components and instruments available in the laboratory.

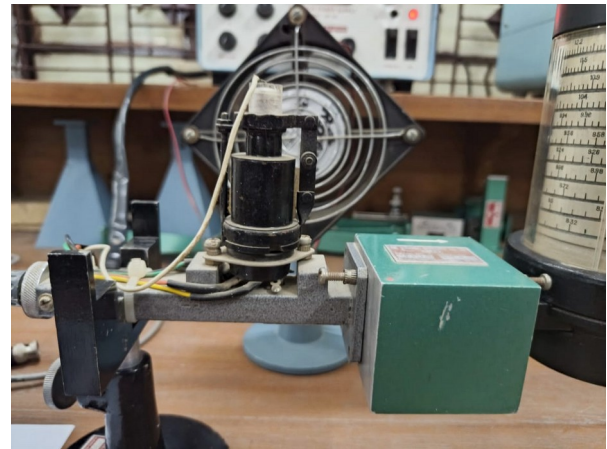
Equipment & Components: Klystron power supply, Gunn power supply, Frequency meter, VSWR meter, Klystron Tube, Gunn diode, PIN modulator, Isolator, E plane Tee, H plane Tee, Directional coupler, Matched termination, movable short, horn antenna etc.

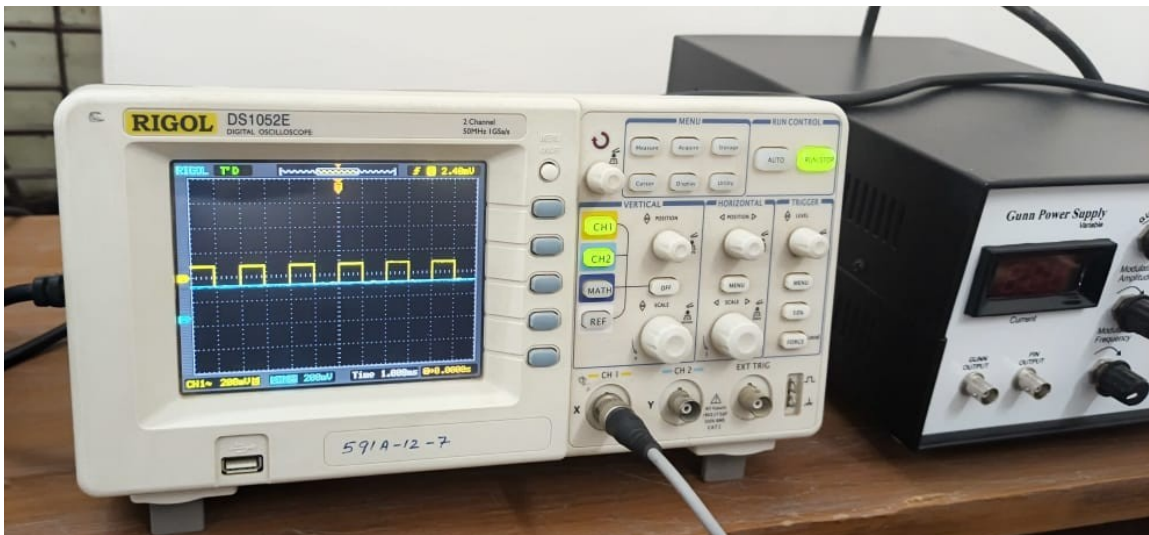
PROCEDURE:

- Note relevant technical specifications of the instruments.
- Study position and functions of the front panel controls of the equipment.
- Know basic principle of operation and functional block diagram of the instrument.
- Facilities provided and limitations of the equipment if any.
- Know initial settings of controls of the equipment before switching on the supply.
- Precautions to be taken while carrying out the measurements.

Observations:

1. Identification through labels is provided on each component.





Digital Signal Oscilloscope (left) and Gunn Power Supply

2. Initial conditions and precautions:

- Do not touch Klystron tube as it may give shock if black wired cap is misplaced.
- Bench setup should be in proper order like all stickers must be on front side.
- Don't try to see the signal inside guide as it may be absorbed by body and may cause some side effects.
- Set micrometer of variable attenuator at 11 mm approx. As height of rectangular waveguide is 10.02mm in this position of micrometer it offers less attenuation.
- Never keep Frequency meter needles closed to lock.
- If we don't require tunable probe remove and cover with cap to protect needle otherwise keep as it is in setup and adjust at proper position.
- Before switching ON klystron power supply make sure Beam voltage knob at min position, Reflector voltage knob at max position.
- Before switching ON VSWR meter make sure gain control knobs at mid position, input switch low impedance and lower position Range selector.
- Before switching OFF, make sure all knob positions remain at their minimum levels.

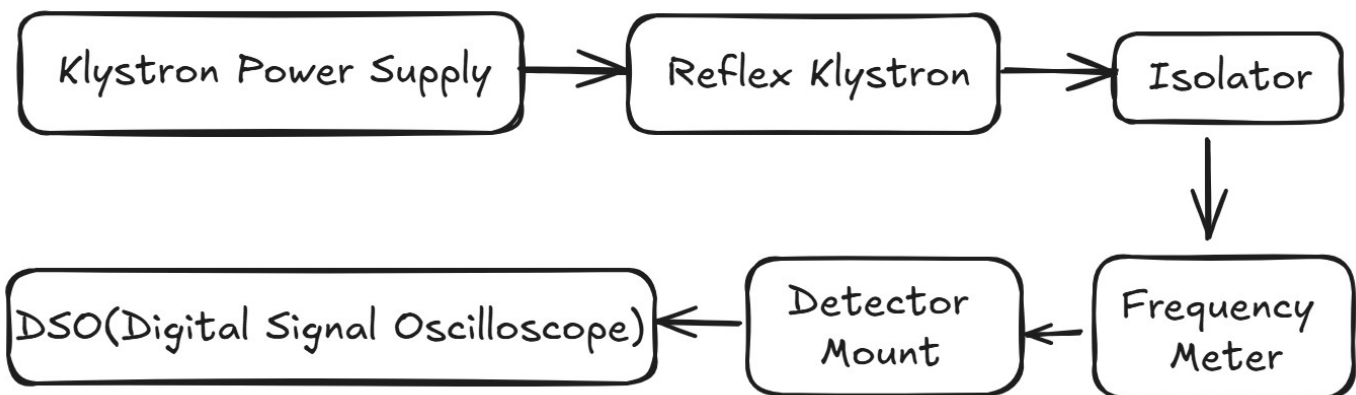
3. Function of Each Equipment/Component:

- **Klystron Tube:** A high-power microwave amplifier that uses electron beam velocity modulation to generate or amplify microwave signals. It serves as a stable, high-frequency source in microwave systems and can produce significant power output for various applications.
- **Klystron Mount:** A mechanical housing that securely holds the klystron tube and provides proper electrical connections to the waveguide system. It ensures optimal coupling between the klystron and the transmission line while maintaining safety and stability.
- **Gunn Oscillator:** A solid-state microwave source that generates continuous wave signals using the Gunn effect in semiconductor materials like gallium arsenide. It provides a compact, reliable alternative to vacuum tube oscillators for lower power applications.
- **PIN Modulator:** Uses PIN diodes to control microwave signal amplitude or provide on/off switching. The modulator allows for precise control of signal power and can create pulsed microwave signals from continuous wave sources.
- **VSWR Meter:** Measures the Voltage Standing Wave Ratio in transmission lines, indicating how well the load is matched to the source. It helps identify impedance mismatches and reflections in the microwave system.
- **Isolator:** A non-reciprocal device that allows microwave energy to pass in only one direction while blocking reflections from reaching the source. This protects sensitive components like oscillators from damage due to reflected power.
- **Variable Attenuator:** Provides adjustable attenuation of microwave signals, allowing precise control of power levels throughout the system. It's essential for calibration and ensuring proper signal levels for measurements.
- **Frequency Meter:** A resonant cavity device used to accurately measure the frequency of microwave signals. It typically uses a calibrated cavity with a tuning mechanism to determine operating frequencies.
- **Slotted Section:** A section of waveguide with a longitudinal slot that allows a probe to sample the electric field. It's used to measure standing wave patterns and determine impedance characteristics of loads and components.
- **Tunable Probe:** A movable probe that can be inserted into the slotted section to detect electric field intensity at different positions. It enables measurement of standing wave patterns and voltage distribution along the transmission line.
- **Detector Mount:** Houses a crystal detector diode that converts microwave energy into a DC signal proportional to the microwave power. This allows measurement and monitoring of microwave power levels using conventional meters.
- **Matched Termination:** Provides a load with characteristic impedance matching the transmission line, absorbing all incident power without reflections. It's used as a reference load and to terminate unused ports.
- **E-plane TEE:** A three-port junction where the branching arm extends in the direction of the electric field. It acts as a power divider/combiner and is commonly used in microwave circuits for signal splitting.
- **H-plane TEE:** A three-port junction where the branching arm extends perpendicular to the electric field direction. It provides different phase relationships compared to E-plane TEE and is used for specific circuit applications.
- **Magic TEE:** Combines both E-plane and H-plane TEE characteristics in a four-port junction. It provides isolation between certain ports and is widely used in balanced mixers, bridges, and antenna feed systems.
- **Fixed Attenuator:** Provides a predetermined, constant level of signal attenuation. It's used for power level adjustment, impedance matching, and protecting sensitive components from excessive power levels.
- **Wave Guide:** A hollow metallic conduit that guides electromagnetic waves from one point to another with minimal loss. It serves as the primary transmission medium for microwave energy in the system.

- **Solid Dielectric Cell:** Contains dielectric material samples for measuring their electrical properties at microwave frequencies. It allows determination of dielectric constant and loss tangent of various materials.
- **M.H.D Coupler (Multi-Hole Directional Coupler):** Samples a portion of the forward or reverse traveling wave without significantly affecting the main signal. It's used for power monitoring and signal injection in microwave systems.
- **T-circulator:** A three-port non-reciprocal device that routes microwave energy in a circular pattern from port to port. It's commonly used in radar systems and provides isolation between transmitter and receiver.
- **Y-circulator:** Similar to T-circulator but with a Y-shaped configuration. It provides the same non-reciprocal routing function but with different physical geometry for specific applications.
- **Phase Shifter:** Introduces a controlled phase delay to microwave signals without significantly affecting amplitude. It's essential for phased array antennas and various microwave measurement applications.
- **Standard Gain Antenna:** A reference antenna with precisely known gain characteristics used for calibration and measurement purposes. It provides a standard for comparing other antennas and measuring field strength.
- **Sectorial Horn – H-plane:** A horn antenna that flares only in the H-plane direction, providing directional radiation with specific beamwidth characteristics. It's used for moderate gain applications and as a feed for larger antennas.
- **Sectorial Horn – E-plane:** Similar to H-plane horn but flares only in the E-plane direction, creating a different radiation pattern. It provides complementary characteristics to H-plane horns for various applications.
- **Pick up Horn:** A receiving antenna designed to intercept and collect microwave energy for measurement purposes. It converts electromagnetic waves back into guided waves for detection and analysis in the measurement system.

Conclusions:

Block Diagram



EXPERIMENT NO. 3

REFLEX KLYSTRON CHARACTERISTICS

Date: 19/8/2025

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Aim: To study the characteristics of reflex klystron.

Equipment & Components: Klystron tube, Klystron power supply, Isolator, Variable attenuator, Frequency meter, Slotted section with Tunable probe, Detector mount, DSO

Theory:

The Reflex Klystron is a microwave tube used as a microwave source in the lab. It makes use of velocity modulation to transform a continuous electron beam into microwave power. Its oscillation frequency can be varied over a wide band and it can be pulse and frequency modulated. Fig. 3.1 shows a schematic of cross-sectional view of a typical reflex klystron tube.

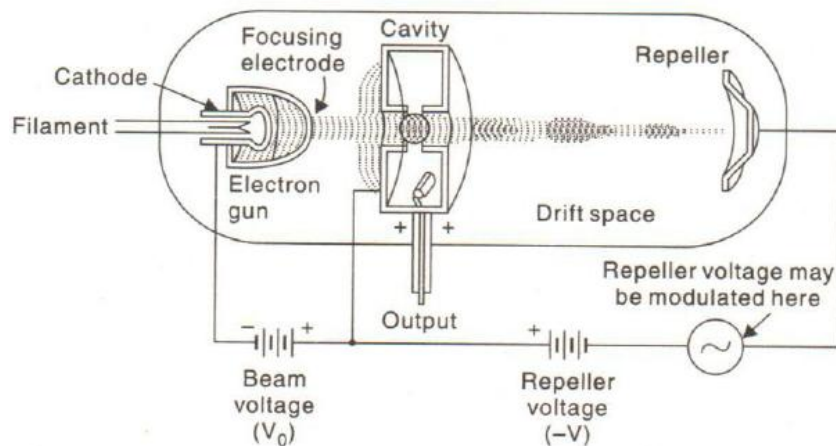


Figure 3.1: Cross-sectional view of a typical reflex klystron tube

Electrons emitted from the cathode are accelerated by and pass through the positive resonator grids towards the reflector. The reflector is at a negative voltage with respect to cathode, and consequently it retards and finally reflects (reflex klystron) the electrons, which then turn back through the resonator grids: In case the klystron starts

to oscillate, a hi-field exists between the resonator grids. The electron traveling through the grid will be either accelerated or retarded as the voltage changes in amplitude. Accelerated electrons leave the grids at an increased velocity and retarded electrons leave at a reduced velocity. Because of the difference in velocity, the electrons leaving the grids will need different time to return (i.e., have different transit times). As a result the returning electron group together in bunches. This Variation in velocity of the electrons is called velocity modulation.

As the electron bunches pass through the resonator grids, they interact with the voltage between the grids. If the bunches pass through the grids at a time such that the electrons are slowed down by the grid voltage, energy will be delivered to the resonator and the klystron will oscillate. Strongest oscillation will occur when the transit time in the reflector resonator region $n + 3/4$ cycles of the resonator frequency, where 'n' is an integer, including zero. If the bunches pass through the grids at a time such that the electrons are accelerated by the grid voltage, energy will be removed from the resonator and no oscillations will occur. Fig. 3.2 shows the variation of output power and frequency with repeller voltage of a typical Klystron tube.

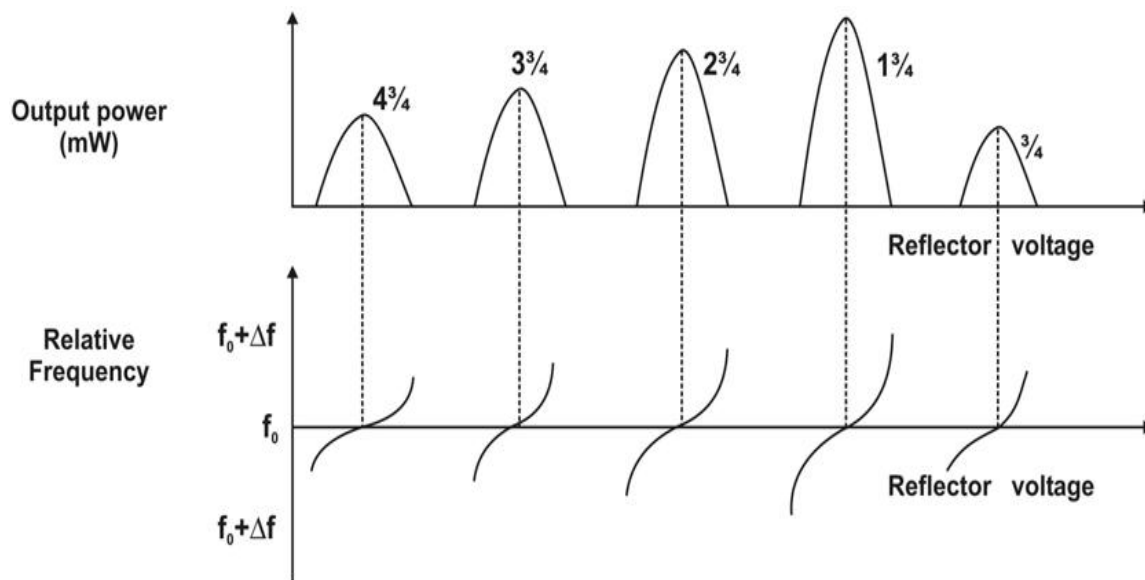


Figure 3.2: Output power and frequency variation with Reflector Voltage

Experimental setup for study of klystron modes:

The Fig.3.3 shows the block diagram of the experimental setup for study of klystron characteristics.

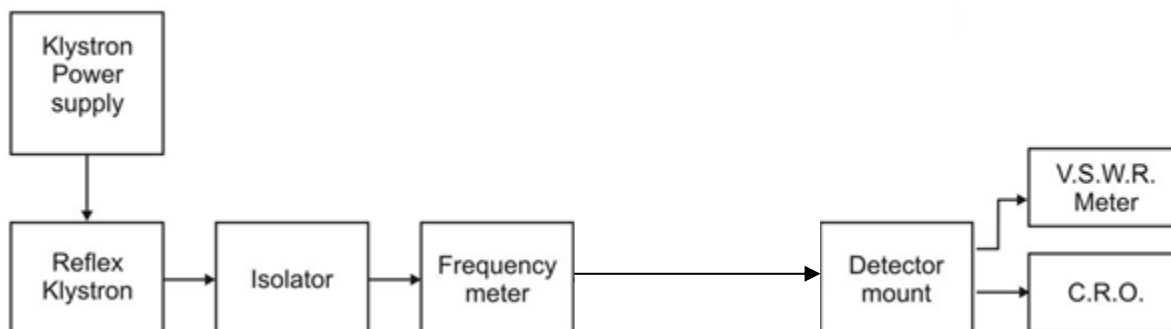


Figure 3.3: Experimental setup for study of Klystron Characteristics

Procedure:

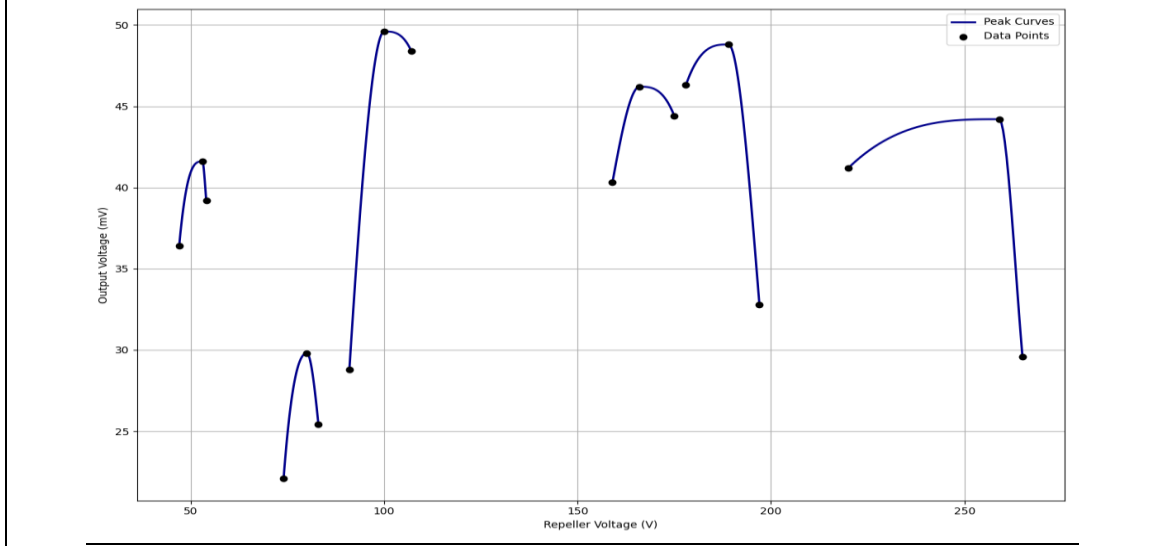
Mode studies:

1. Connect the components and equipment as shown in experimental setup.
2. Rotate the frequency meter at one side.
3. Switch on the klystron power supply, DSO and cooling fan for the Klystron tube. Wait for 1-2 minutes for the klystron to respond.
4. Cathode voltage knob at minimum position gives a beam voltage of 235 V. Vary beam voltage up to 300V, keep repeller voltage maximum and Observe beam current on the meter by changing meter switch to beam current position. "The beam current should not be more than 30mA".
5. Now change the meter switch to repeller voltage position.
6. To measure frequency, switch the Mode-switch of Klystron to AM mode and observe output on DSO display. Use AM amplitude, frequency controls and controls on Oscilloscope front panel, try to get clear display on CRO.
7. By varying repeller voltage observe output voltage variation in the DSO and the modes.
8. For each repeller voltage, measure the output frequency using Direct Frequency Reading Meter (DFRM). For measuring the frequency, rotate the drum of DFRM such that the output voltage on the DSO is minimum and note the direct reading on the circular scale of DFRM.
9. Plot two graphs (i) output voltage observed on the DSO versus the repeller voltage and (ii) frequency versus the repeller voltage.
10. Observe the number of modes of the Klystron.

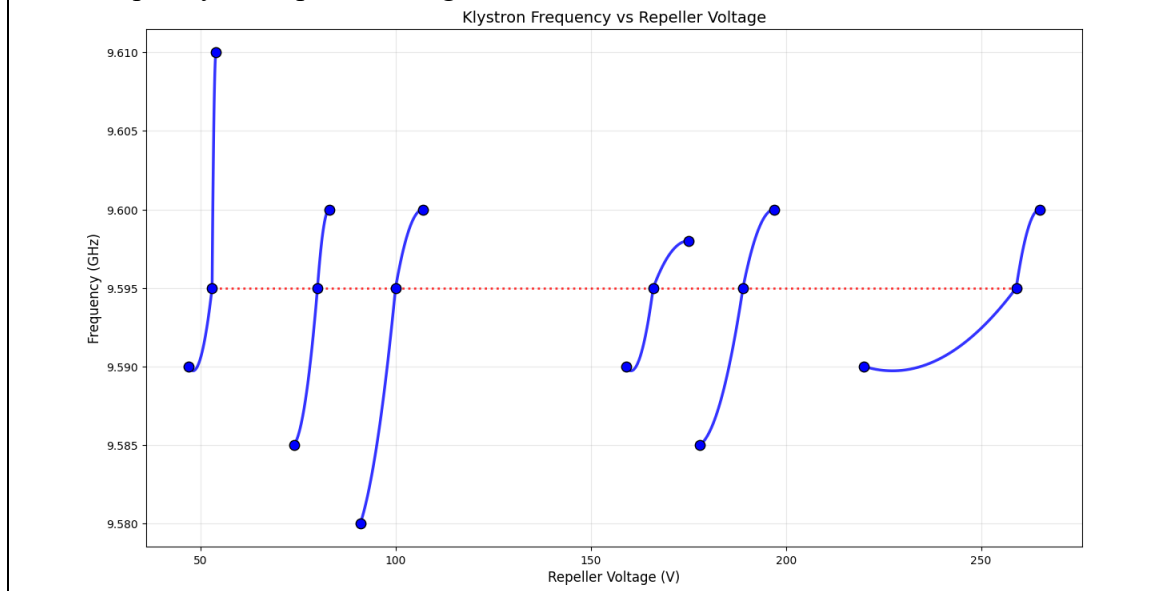
Observations:**Mode studies**

Sl. No.	Repeller voltage (V)	Output Voltage (V)	Frequency (GHz)
1	265	29.6	9.6
2	259	44.2	9.595
3	220	41.2	9.59
4	197	32.8	9.6
5	189	48.8	9.595
6	178	46.3	9.585
7	175	44.4	9.598
8	166	46.2	9.595
9	159	40.3	9.59
10	107	48.4	9.6
11	100	49.6	9.595
12	91	28.8	9.58
13	83	25.4	9.6
14	80	29.8	9.59
15	74	22.1	9.585
16	54	39.2	9.61
17	53	41.6	9.595
18	47	36.4	9.59

Plot Output voltage vs Repeller voltage



Plot Frequency vs Repeller voltage



Conclusion:

From the graphs we can observe that the reflex klystron operates in several **modes** as the repeller voltage is varied. At certain repeller voltages, the output voltage reaches a maximum, indicating strong oscillations, while in between these regions the output falls, showing mode separation. The oscillation frequency remains nearly constant around **9.595 GHz**, since it is primarily determined by the resonant cavity. Thus, the experiment verifies that the reflex klystron generates microwave power through **velocity modulation and electron bunching**, and that its output strongly depends on the repeller voltage, with distinct modes of operation.

EXPERIMENT NO. 5

VSWR MEASUREMENT

Date: _____ PS No.: _____ Batch No. _____

ID No. _____ Name: _____

Aim: To become familiar with the basic technique for measuring Voltage Standing Wave Ratio (VSWR).

Equipment & Components: Klystron power supply, Klystron mount, Klystron tube, Isolator, Frequency meter, Variable attenuator, Slotted section, Tunable probe, wave guide stands, Movable short Load, matched termination, BNC cable, SS tuner, VSWR meter.

Theory:

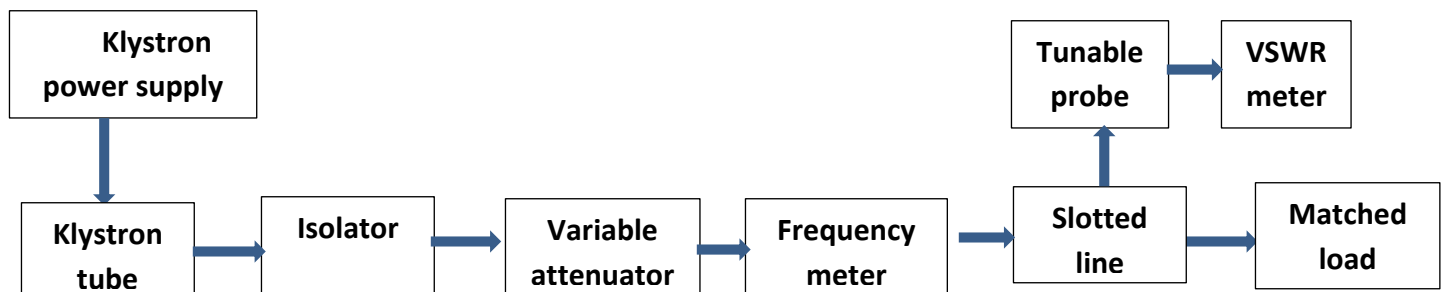
The electromagnetic field at any point of transmission line or waveguide may be considered as the sum of two traveling waves: the 'incident wave' propagates from generator and the 'reflected wave' propagates towards the generator. The reflected wave is set up by reflection of incident wave from a discontinuity on the line or from the load impedance. The magnitude and phase of reflected wave depends upon amplitude and phase of the reflecting impedance. The presence of two traveling waves gives rise to standing wave along the line. The maximum field strength is found where two waves are in phase and minimum where they are in opposite phase. The distance between two successive minimum (or maximum) is half the guided wave-length on the line. The ratio of electrical field strength of reflected and incident wave is called reflection coefficient. The VSWR is defined as ratio between maximum and minimum field strength along the line. Hence, VSWR is

$$\text{VSWR} = E_{\max} / E_{\min} = (E_i + E_r) / (E_i - E_r) \text{ where } E_i = \text{incident voltage and } E_r = \text{reflected voltage}$$
$$= (1 + \text{reflection co-efficient}) / (1 - \text{reflection co-efficient})$$

Reflection Co-efficient (Γ) the size of reflection $\Gamma = (\text{VSWR} - 1) / (\text{VSWR} + 1)$

If a matched termination is connected to the transmission line, the reflected wave is zero, the reflection co-efficient is also zero, and the VSWR is 1.0. If reflection co-efficient is one, there is total reflection. It should be noted that reflection co-efficient must lie between zero and one. Reflection co-efficient (R) = the ratio of voltage reflected (E_r) to the voltage incident (E_i). The voltage standing wave ratio is defined as ratio between maximum and minimum field strength along the line.

EXPERIMENTAL SETUP:



PROCEDURE:

Measurement using VSWR meter

1. Set the equipment as shown in experimental setup.
2. Keep variable attenuator in the minimum attenuation position (attenuator micrometer reading above 11 mm).
3. Keep the control knob of VSWR meter as below,

Range dB	-	40db to 50db
Input switch	-	Low impedance
Meter switch	-	Normal
Gain	-	Mid position
4. Keep the control knob of Klystron power supply as follows:

Beam voltage	-	Off
Mod-switch	-	Am
Beam voltage knob	-	Fully anti clockwise
Reflector voltage knob	-	Fully clockwise
AM-amplitude knob	-	Fully clockwise
AM frequency & amplitude knob	-	Mid position
5. Switch On the klystron power supply, VSWR meter & cooling fan
6. Switch On the beam voltage switch and set beam voltage at 300V
7. Observe square waveform in DSO by varying reflector voltage and connect probe to VSWR meter
8. The probe is moved along the slotted line for maximum deflection in the VSWR meter.
9. The gain control knob of the VSWR meter or the variable attenuator is adjusted until the meter indicates 1.00 on normal SWR scale
10. Measure the frequency by using the direct reading frequency meter (DRFM)
11. Keeping all control knobs as it is, the probe is adjusted for minimum position on the SWR reading. The VSWR meter reading is recorded (SWR meter reading at the minimum, when the maximum is set for 1.00, gives directly the VSWR). Note that SWR NORMAL scale is for $1.00 \leq s \leq 4.00$ and EXPANDED scale is for $1.00 \leq s \leq 1.33$.
12. If the VSWR is between 3.20 and 10.00, the range dB switch is changed to next higher position and the VSWR reading is taken by the VSWR scale of 3.00 to 10.00 (second SWR NORMAL scale).
13. Remove the matched termination and repeat steps 8-10 for getting VSWR for open circuit, short circuit and unknown load.
14. Repeat the above steps from 8 to 12 at five different repeller voltages.
15. Plot the variation of VSWR of the unknown load with frequency.

Measuring VSWR by using DSO:

1. Connect the tunable probe to DSO, set some reflector voltage and observe square waveform in DSO.
2. Move the tunable probe along slotted line and note down maximum and minimum amplitudes in DSO.
3. Calculate VSWR by using formula, $VSWR = V_{max} / V_{min}$
4. Calculate Reflection coefficient $\Gamma = (VSWR - 1) / (VSWR + 1)$
5. Repeat above steps at different loads.

OBSERVATIONS:**Table for VSWR measurement using VSWR meter**

S. No.	Repeller Voltage (V)	Frequency f_o	Load –end matched termination	Load-end open-circuited	Load- end short-circuited	Unknown load (Movable Short)
1						
2						
3						
4						
5						

Plot variation of VSWR with frequency for unknown load**VSWR measurement using DSO (Beam voltage: 277V, Beam current 19mA, Repeller Voltage: 240 V)**

S.No	Load	$V_{\max}$ (p-p) (Volts)	$V_{\min}$ (p-p) (Volts)	VSWR(s) $V_{\max}/V_{\min}$	$\Gamma = (s-1) / (s + 1)$
1					
2					
3					
4					

CONCLUSIONS:

EXPERIMENT NO. 4

FREQUENCY MEASUREMENT

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Aim: To determine the frequency and wavelength in a rectangular waveguide working in TE₁₀ mode.

Equipment & Components: Klystron Power Supply, Klystron Tube 2k25 , Klystron Mount, Isolator, Frequency Meter, Variable Attenuator, Detector Mount, Wave Guide Stands, DSO., BNC Cables.

Theory:

The measurement of frequency or wavelength is one of the primary requirements in most microwave measurements. Frequency is the most fundamental quantity because the frequency of oscillation is the same at all parts of the microwave system under steady - state conditions. Wavelength on the other hand depends upon the configuration of the electric and magnetic fields as determined by the geometry of the measuring device. The relation between length of the transmission line, frequency and wavelength terminated in a short circuit are reviewed briefly as shown in experimental setup. An input voltage wave of the quarter - wave line at 'a' will have phase change of 90 degree in traveling from the input (open) to the shorted end, 180 degree phase change at the short, and another 90 degree in relating to the open end . The total phase change is 360 degree and the reflected wave is in phase with the applied wave. The different voltage and current pattern are referred to as modes.

The resonant mode excited by the possible frequency is called the fundamental mode or the dominant mode. A better way to measure frequency is with a calibrated resonant cavity. A resonant cavity wave meter is the analog of a tuned resonant circuit, in general there are two primary types (1) Transmission cavities, which pass only the signal to which they are tuned and (2) Absorption cavities , which attenuate (by absorption) only the frequency to which they are tuned . An absorption type is preferred for laboratory frequency measurements. First, the power level is adjusted to give a full scale reading on the output VSWR meter then the wave meter is tuned slowly until there is a dip in the power level. The frequency may then be read from the dial of the meter.

For dominant TE₁₀ mode in rectangular wave guide λ_0 , λ_g and λ_c are related as follows:

$$\frac{1}{\lambda_o^2} = \frac{1}{\lambda_g^2} + \frac{1}{\lambda_c^2}$$

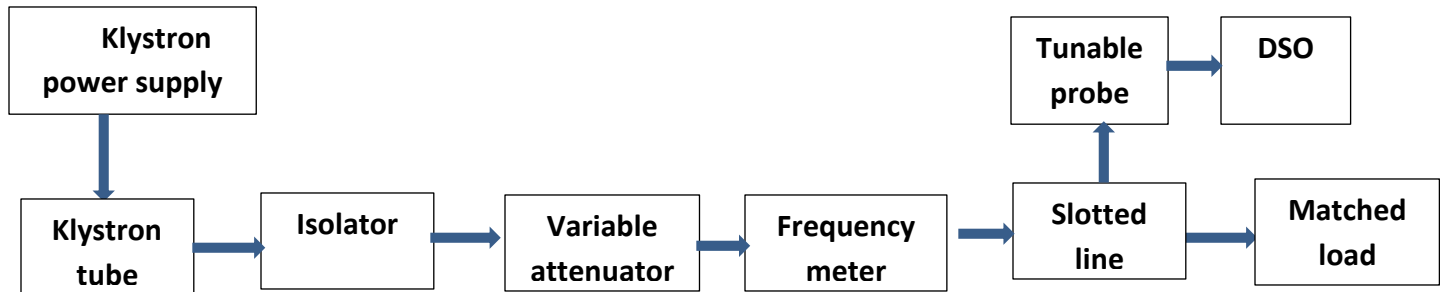
Where, λ_o is free space wavelength

λ_g is guide wavelength

λ_c is cutoff wavelength

For TE₁₀ mode, $\lambda_c = 2a$ where 'a' is inner broad dimension of waveguide cross-section.

EXPERIMENTAL SETUP:



PROCEDURE:

- Connect the components and equipment as shown in experimental setup.
- Keep the Variable Attenuator to minimum attenuation position.
- Keep the control knob of Klystron power supply as below:
 - Mode switch : AM
 - Beam voltage knob : Fully anti-clockwise
 - Repeller voltage knob : Fully clockwise
 - Meter switch : Beam Voltage position
- Rotate the frequency meter at one side.
- Switch on the Cooling fan, Klystron power supply and DSO.
- Beam voltage knob at minimum position gives a beam voltage of 235 V.
- Increase the Beam voltage up to 300V and wait for 1-2 minutes to respond the Klystron tube then reduce the Beam voltage up to 265V.
- Now change the meter switch to Repeller voltage position. Decrease the reflector voltage to get proper square wave output on DSO. Once the square wave is observed, adjust the amplitude knob in AM section and tunable probe of slotted section for maximum amplitude of square wave. Note down the Repeller voltage & output voltage.
- This is basic set up of Klystron tube for normal operation.
- To measure frequency, rotate the frequency meter and observe for dip in the output and note the corresponding frequency.
- Move probe along the slotted section and observe maxima & minima amplitude on DSO. Measure the distance between two successive maxima or minima by scale provided on slotted section and calculate guide wavelength (λ_g).
- Measure the wave guide inner broad dimension 'a' which will be around 2.286 cm for X band. Calculate cut-off wavelength (λ_c).
- Calculate free space frequency (f_o), & verify with frequency obtained by frequency meter.
- Verify for five different frequencies.

FORMULA:

- 1) To calculate guide wavelength (λ_g),

$$\frac{\lambda_g}{2} = (d_1 - d_2)$$

Where, ($d_1 - d_2$) is the distance between two successive maxima or minima in cm.

d_1 or d_2 = MSR+VSR(LC) where MSR is Main Scale Reading, VSR is Vernier Scale Reading (VSR) and LC is least count.

- 2) To calculate cut-off wavelength of waveguide (λ_c),

$$\lambda_c = 2a \text{ (in cm)}$$

Where, 'a' is the inner broad dimension of waveguide cross-section.

- 3) To calculate free space frequency (f_o),

$$f_o = \frac{c}{\lambda_o} = c \sqrt{\frac{1}{\lambda_g^2} + \frac{1}{\lambda_c^2}}$$

Where, $C = 3 \times 10^{10}$ cm/sec. i.e. Velocity of light in free space

OBSERVATIONS:

S.No.	Repeller Voltage (volts)	Direct Frequency Meter Reading (GHz)	Guided Wavelength (cm)			λ_o (Free Space Wavelength in cm)	f_o (Frequency in GHz)
			d_1	d_2	λ_g		
1.	269V	9.595	11.34	13.25	3.82	2.93	10.20

CONCLUSIONS:

The wavelength of the guided wave was calculated to be 3.82 cm, from which the open air wavelength was calculated to be 2.93 cm. The frequency was then determined to be 10.20 Ghz, which was compared with the observation from the frequency meter.

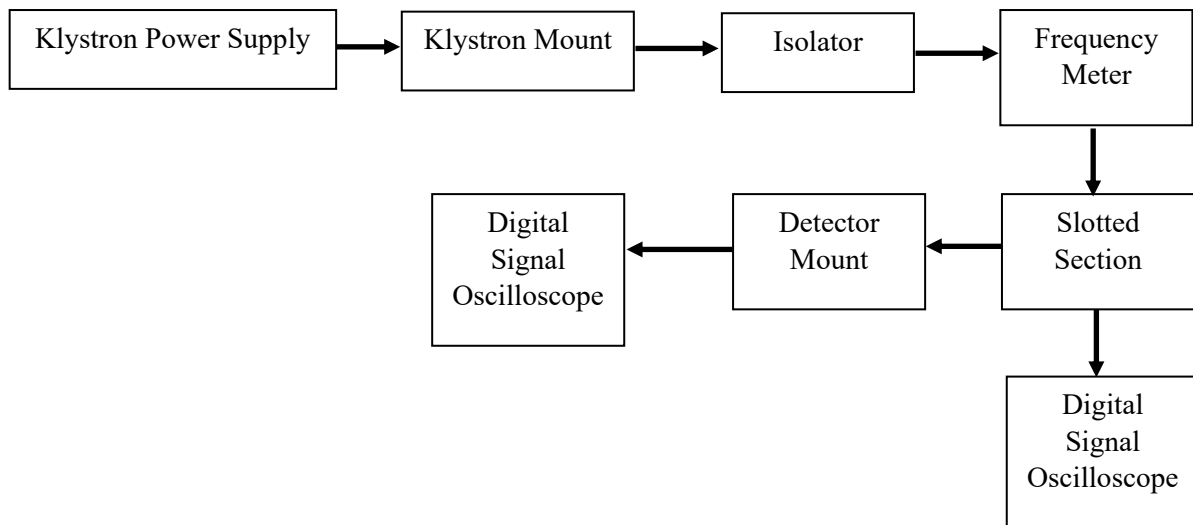
Experiment No. 5

Measurement of the dielectric constant of the given solid.

Date:	Group No.
Name:	ID No:
Name:	ID No:
Name:	ID No:
Name:	ID No:
Name:	ID No:

Objective: Measurement of the dielectric constant of the given solid.

Block Diagram:



Initial Klystron Power Supply:

1. Beam voltage: Minimum
2. Repeller voltage: Maximum
3. Current: 11milliamp
4. Frequency: 9.595GHz
5. Modulation: AM
6. Dielectric sample (solid): Softwood
7. Dimension of the sample: $1.1\text{cm} \times 2.3\text{cm} \times 6.3\text{cm}$ (L $\times$ W $\times$ H)
8. Waveguide dimension: $1.1\text{cm} \times 2.3\text{cm}$ (L $\times$ W)

Procedure & calculation:

1. First minima without dielectric material (d_1) =
2. Second minima without dielectric material (d_2) =
3. Guided wavelength (λ_g) = $2(d_2 - d_1)$
4. Cut-off wavelength (λ_o) = $2a$ [a is the width of the waveguide; 2.3cm]
5. Again, measure first minima with the dielectric solid inside the waveguide:
6. Difference between the first minima (Δs) = $d_3 - d_1$
7. Now calculate the X'' ,

$$X'' = \left(\frac{\lambda_g}{H}\right) \tan (2\pi (\Delta s + H))/\lambda_g$$

$$\left(\frac{1}{\lambda_o}\right)^2 = \left(\frac{1}{\lambda_g}\right)^2 + \left(\frac{1}{\lambda_c}\right)^2 \quad [\text{Determining the frequency}]$$

8. Once the value of the X'' is determined,

$$X'' = \tan(x)/x$$

Keeping the X'' as the y-axis of the $\tan(x)/x$ graph.

9. Through this, we can get ourselves the value for the normalized phase shift (V) in the dielectric measurement process.

$$X = 2\pi V$$

10. To calculate the dielectric constant (ϵ_r)

$$\epsilon_r = 1 - \left(\frac{\lambda_o}{\lambda_g}\right)^2 + \left(\frac{\lambda_o V}{H}\right)^2$$

11. Due to the insertion of the softwood as substrate inside the waveguide, the **attenuation** can also be experienced while fetching the output from the detector mount.

Result & discussion

EXPERIMENT NO. 2

Design and Analysis of a Microstrip Line of characteristic impedance of 50Ω

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Aim: To design a microstrip line of characteristic impedance of 50Ω in CST Studio software

- To plot scattering parameters, E-fields, H fields and visualize the Smith Chart

Equipment & Components: CST Studio

Theory:

Microstrip Line:

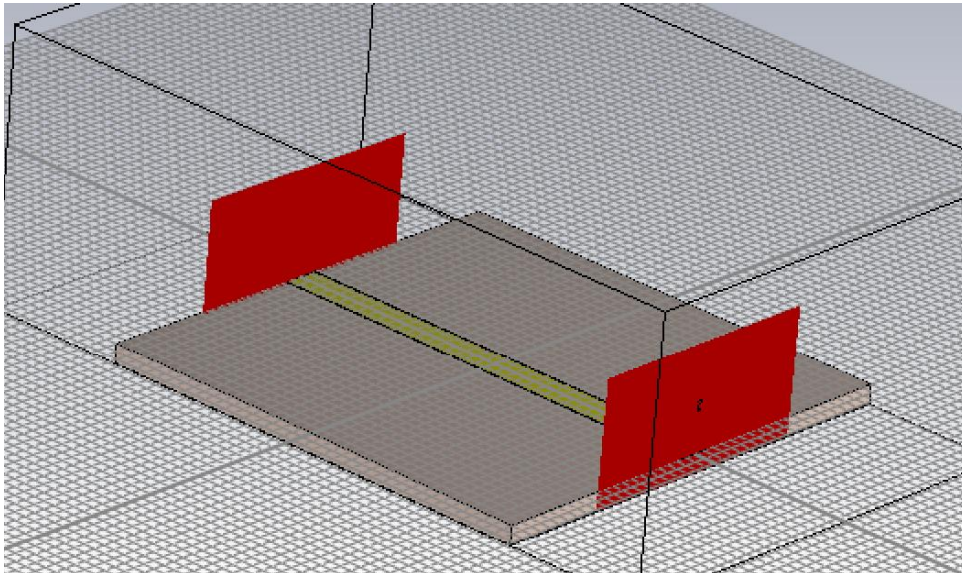


Figure 1 – Microstrip line as setup in CST with waveguide ports setup

A microstrip line is a basic form of a transmission line, constructed by placing a dielectric between a copper sheet and a strip. The copper sheet is connected to ground whereas the excitation is fed to the strip. The characteristics of this transmission line is determined by the width of the strip and the height and permittivity of the dielectric between the copper layers. By varying the width of the strip, we can impart different characteristic impedances to the transmission line.

Design Problem:

For the given experiment, we use a dielectric FR-4, with a dielectric constant of 4.3. The dielectric thickness is taken to be 1.6 mm. We then add a copper layer underneath, which acts as ground with thickness of 0.035 mm. Finally, we add the microstrip, which is also made of copper, above the dielectric with thickness of 0.035 mm. To achieve the 50-ohm characteristic impedance for this experiment, we calculate the width of this strip to be about 3.12 mm.

Observations:

S-Parameters

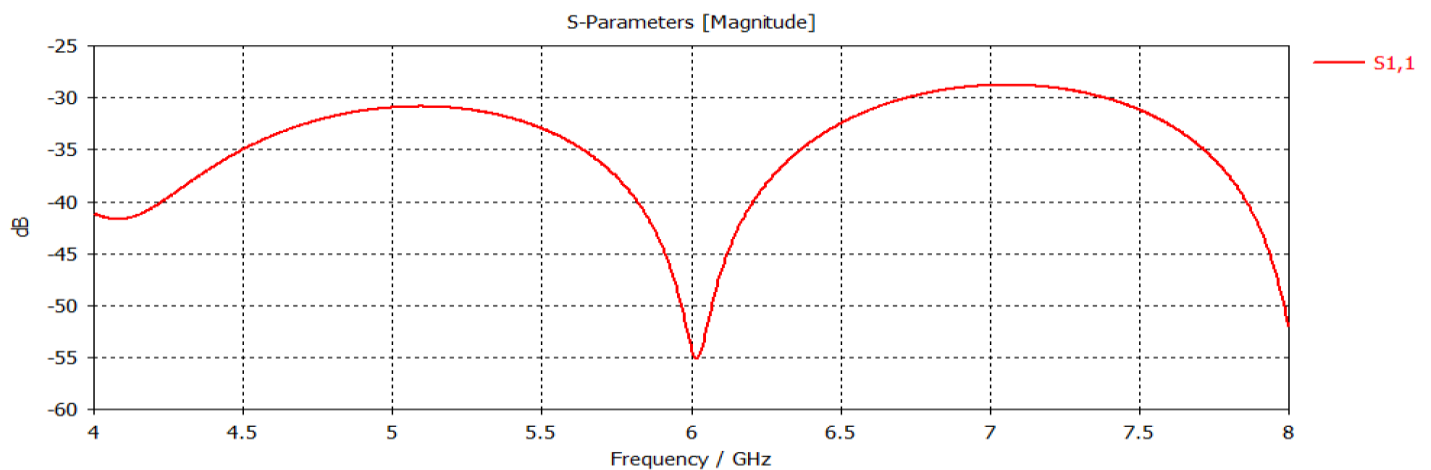


Figure 2 - S11

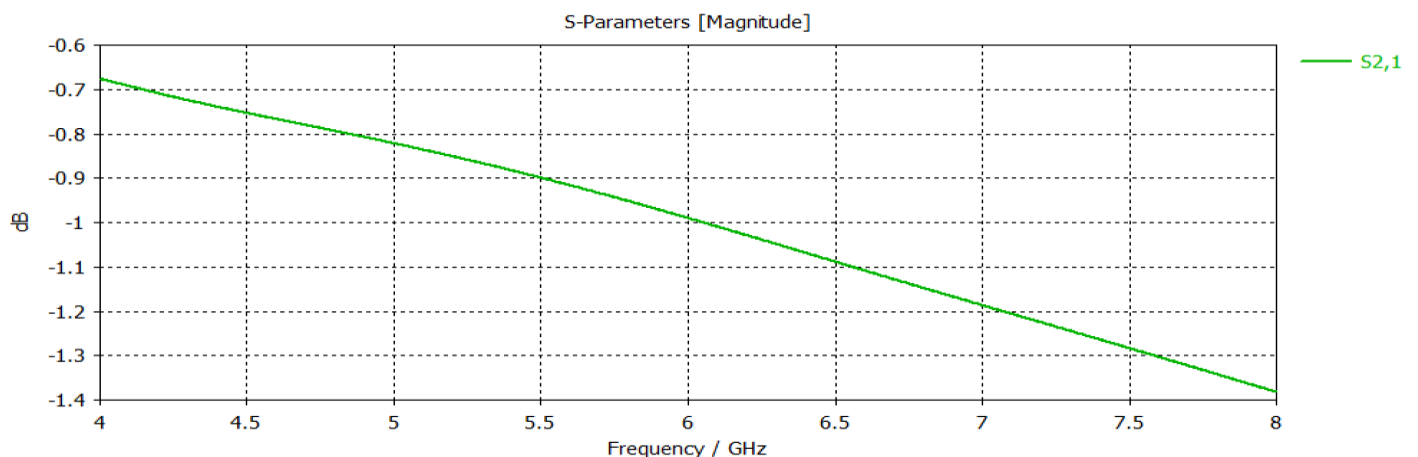


Figure 3 – S21

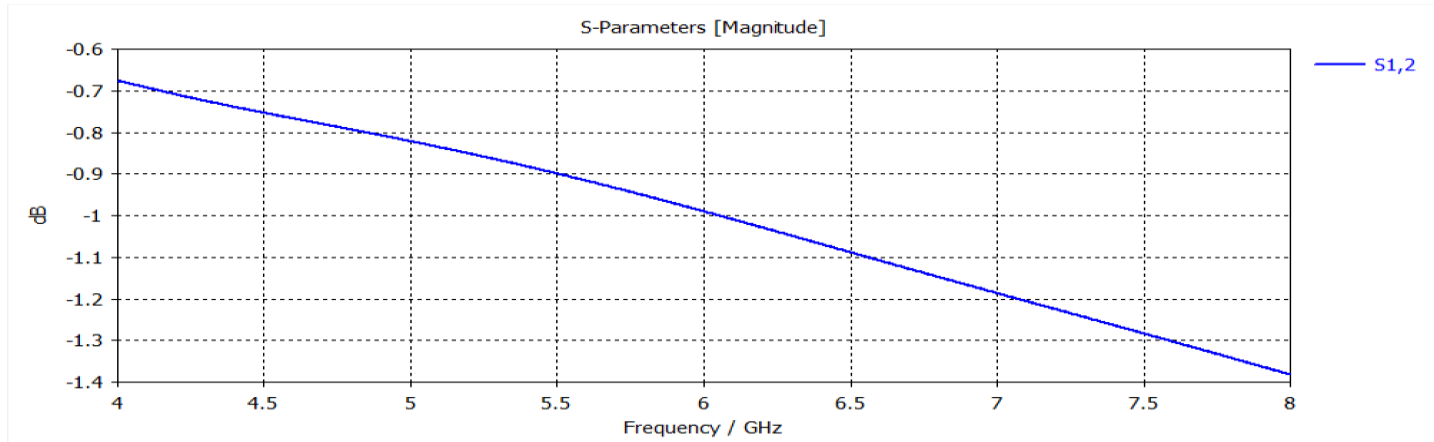


Figure 4 – S21

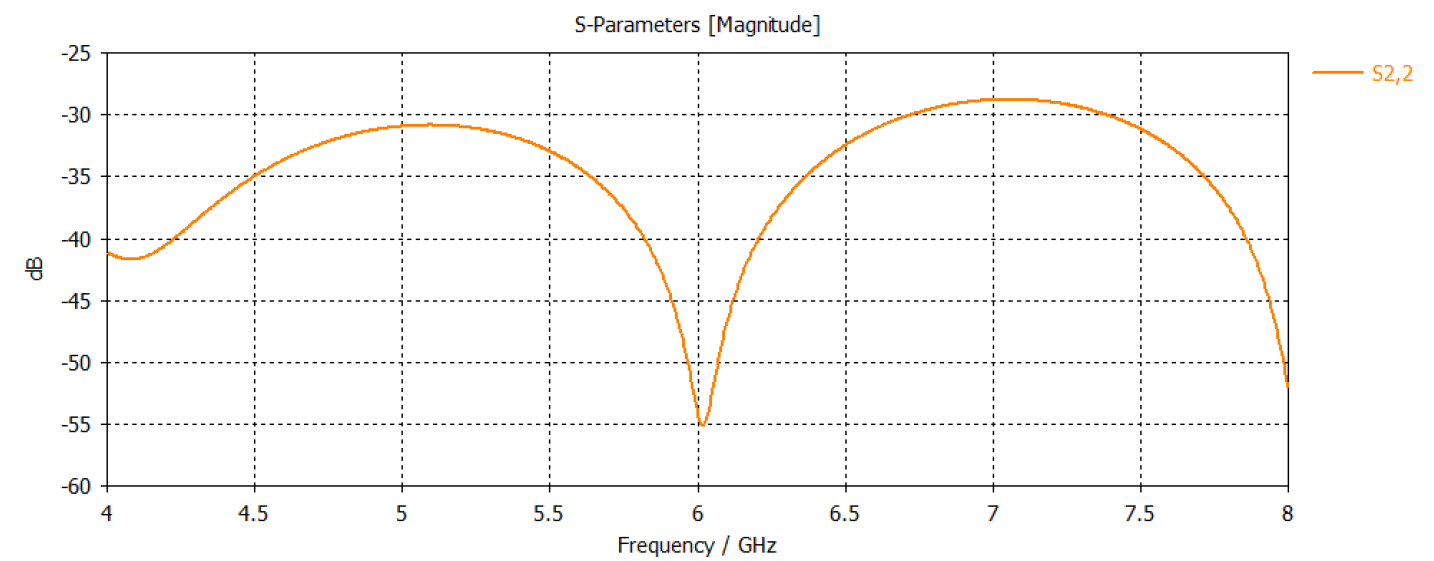
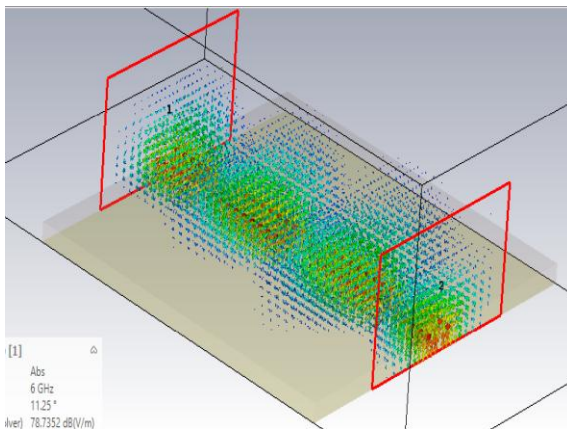
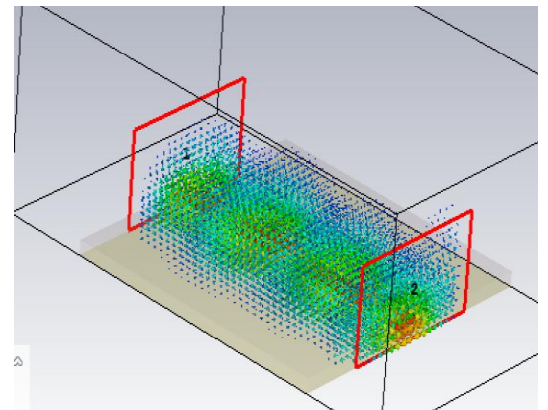


Figure 5 – S22

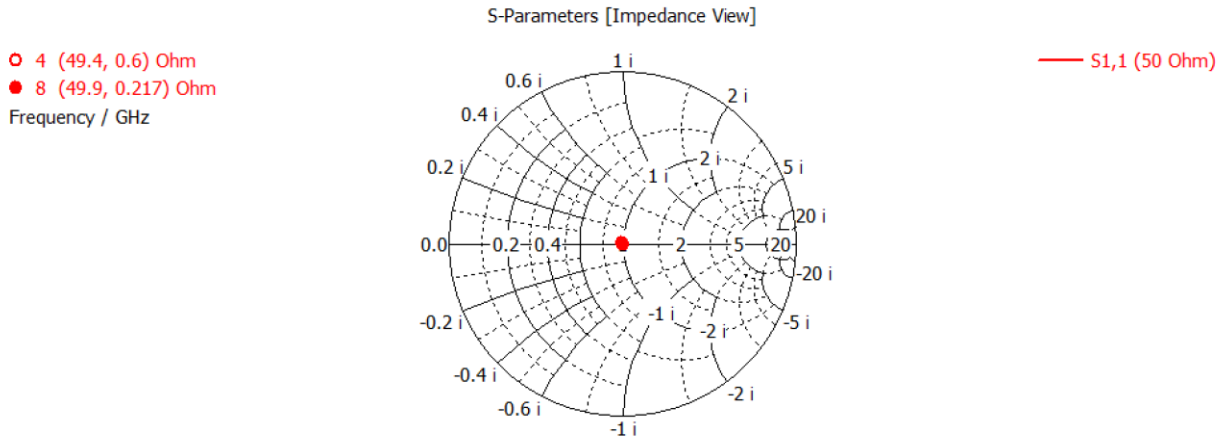
(i) E-field



(ii) H- field



(iii) Smith Chart



Conclusions:

Firstly, we note that since this microstrip transmission line is symmetric we observe the same S_{11} and S_{22} graphs, indicating that the behavior of the transmission line does not depend on which port you pass the excitation to.

Looking at S_{22} graph, we see that this transmission line has a resonant frequency of around 6 GHz. Overall, the values of S_{22} entirely lie below the -10dB threshold, indicating that very less power is being reflected back. If we do the power analysis of S_{22} we observe different amounts of power reflected and transmitted for various frequencies. We take the examples of the resonant frequency and the values of the frequencies where the reflection is highest.

Frequency	S_{22} (dB)	Reflected Power	S_{21} (dB)	Transmitted Power	Lost Power
5.1 GHz	-31 dB	0.08%	-0.85 dB	82.22%	17.70%
6.0 GHz	-55 dB	~0.000%	-1.00 dB	79.43%	20.57%
7.1 GHz	-28 dB	0.16%	-1.25 dB	74.99%	24.85%

We can clearly see that despite fairly low reflected power values, the amount of power transmitted isn't that great. This is primarily because we picked a dielectric that is fairly lossy and that ends up consuming a bunch of the power. We can see clearly that the amount of power lost increases with increasing frequency from both this table and the S_{21} graph.

Zooming in on the E-Field, we observe that the direction of the field is from the strip to the ground plane which is as it should be. We can also see the pulse nature of the electric field across the length of the microstrip. Animating it shows the pulses travelling towards the destination port. Similar observations are made for the H-field

The Smith Chart shows us the impedance value being close to the centre of the circle, indicating a normalized impedance of 1, which is as expected.

EXPERIMENT NO. 3

Defining port impedance of microstrip line with lumped component: Real and complex

Date: 26/2/2025

Group No. 2

Name: Mihika Mundhada

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Name: Pramit Pal

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Name: Saikrishna Bulusu

ID No: 2023AAPS0728P

Name: Somesh Huljute

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Name:

ID No:

Aim: To terminate the 50 Ω transmission line with a lumped element

- To plot scattering parameters and measure the mismatch loss from **experiment 2** for these cases:
 - purely real (50 Ω)
 - purely real (100 Ω)
 - purely complex (inductive and capacitive)
 - combination of real and complex

Equipment & Components: CST Studio

Theory:

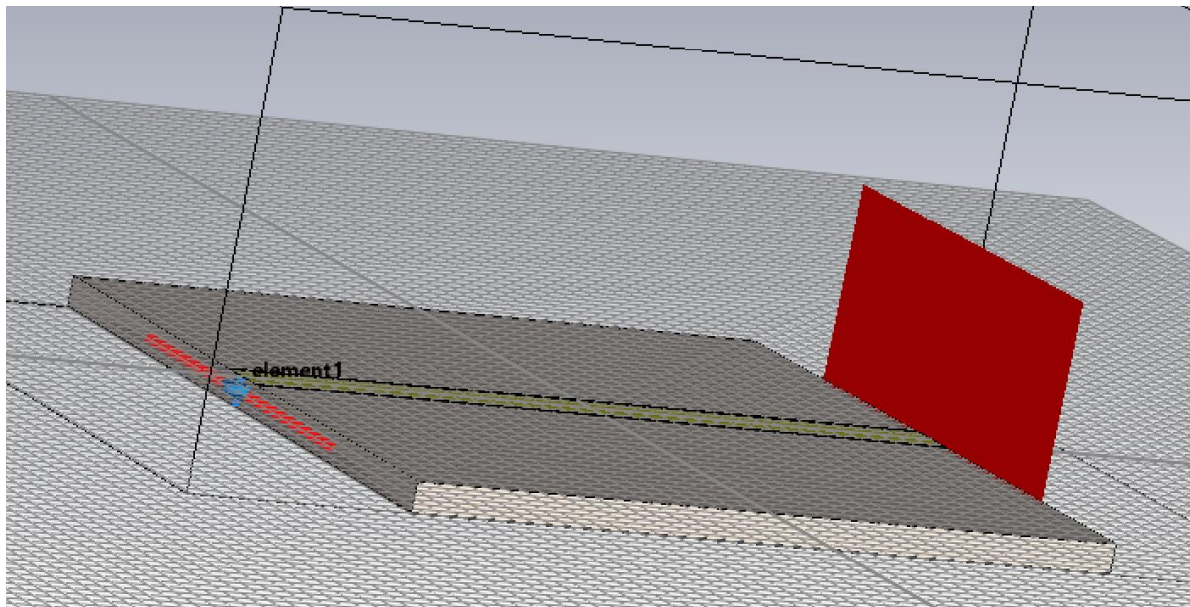


Figure 1 – Microstrip line as setup in CST with one waveguide port and one lumped element

A microstrip line is a basic form of a transmission line, constructed by placing a dielectric between a copper sheet and a strip. The copper sheet is connected to ground whereas the excitation is fed to the strip. The characteristics of this transmission line is determined by the width of the strip and the height and permittivity of the dielectric between the copper layers. By varying the width of the strip, we can impart different characteristic impedances to the transmission line.

When terminating a transmission line, we need to care about additional factor like matching the load impedance to the characteristic impedance of the line. Of particular importance, especially in context to S_{11} , is how the load impedance affects the reflectance of the transmission line. We have,

$$\Gamma = \frac{Z_L - Z_o}{Z_L + Z_o} = 10^{\frac{S_{11}}{20}}$$

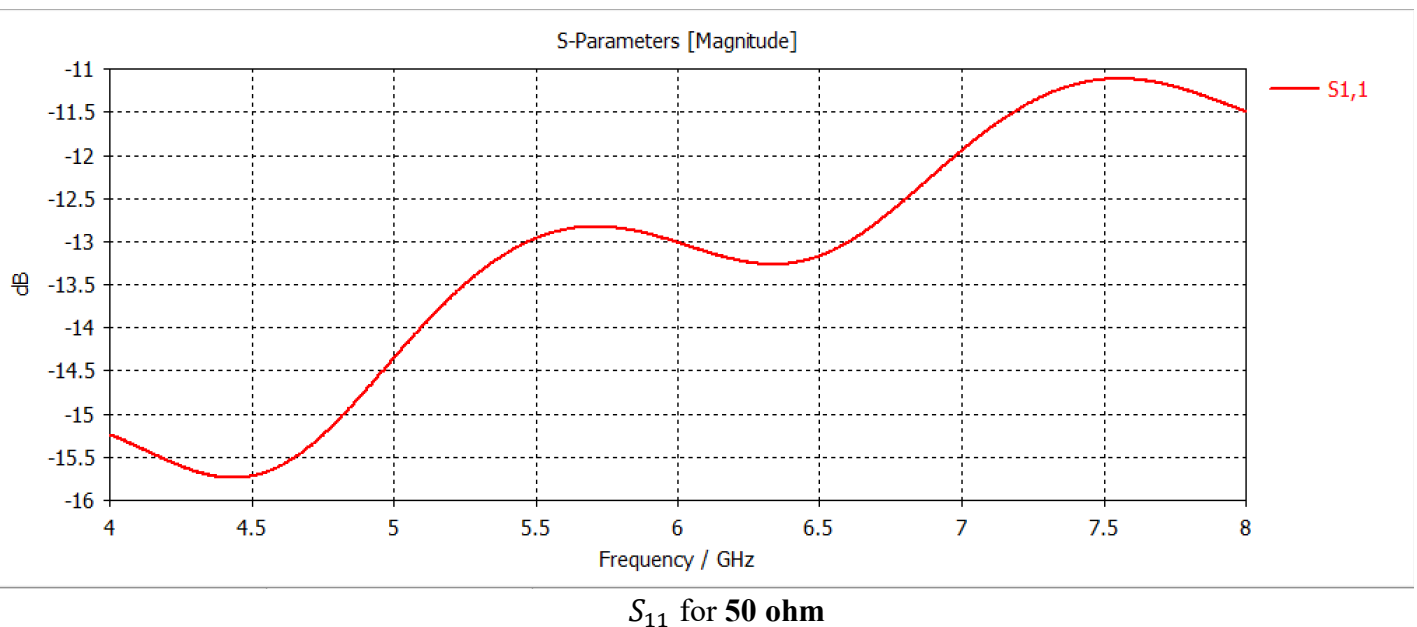
This clearly shows that the reflectance is zero when the load is matched to the transmission line and else, we get larger values.

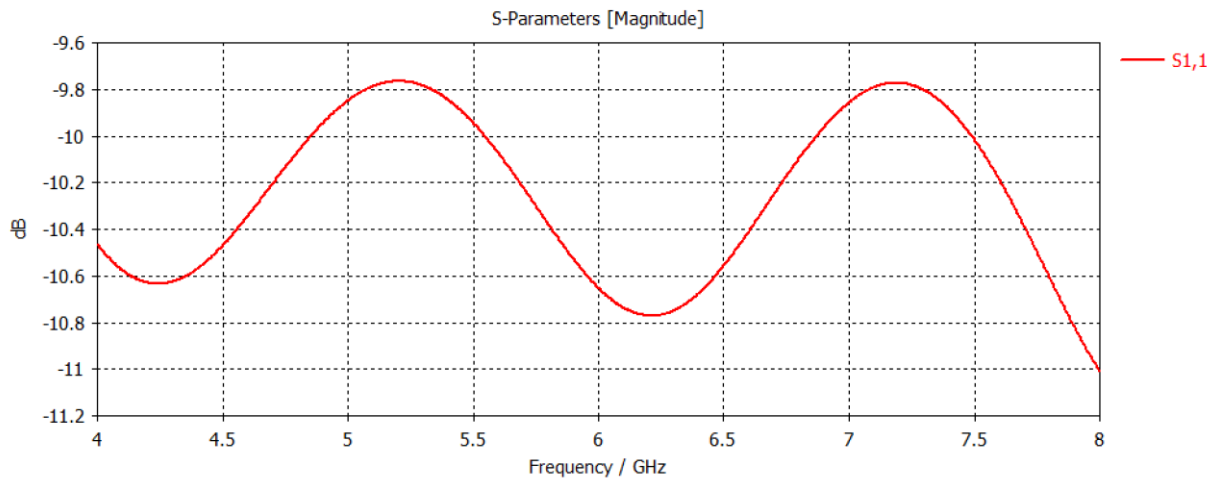
Design Problem:

For the given experiment, we use a dielectric FR-4, with a dielectric constant of 4.3. The dielectric thickness is taken to be 1.6 mm. We then add a copper layer underneath, which acts as ground with thickness of 0.035 mm. Finally, we add the microstrip, which is also made of copper, above the dielectric with thickness of 0.035 mm. To achieve the 50-ohm characteristic impedance for this experiment, we calculate the width of this strip to be about 3.12 mm.

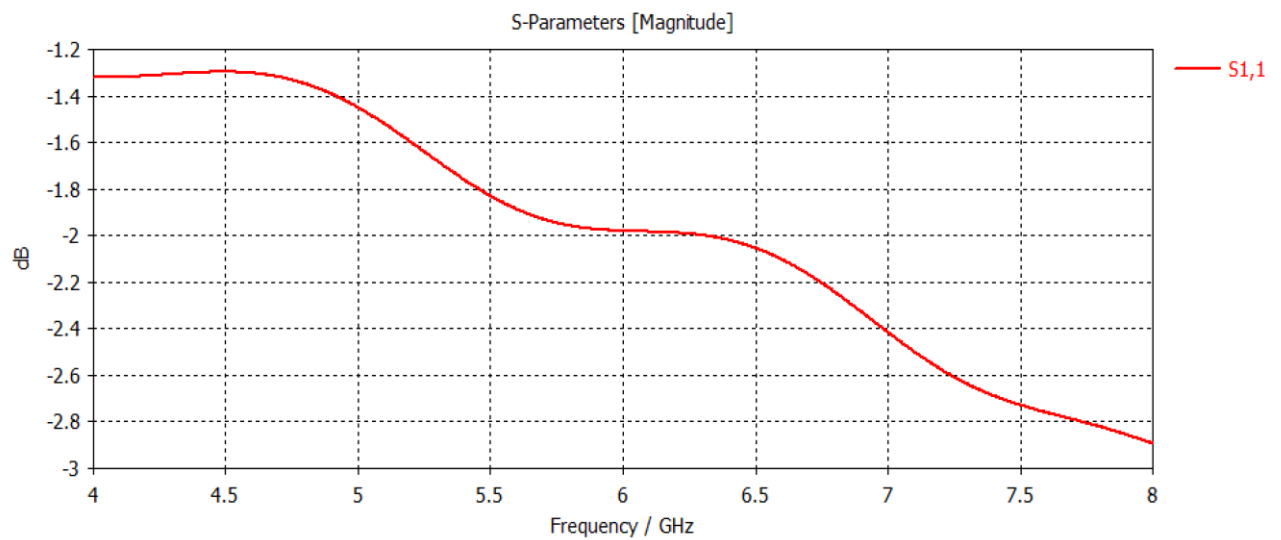
Observations:

S-Parameters

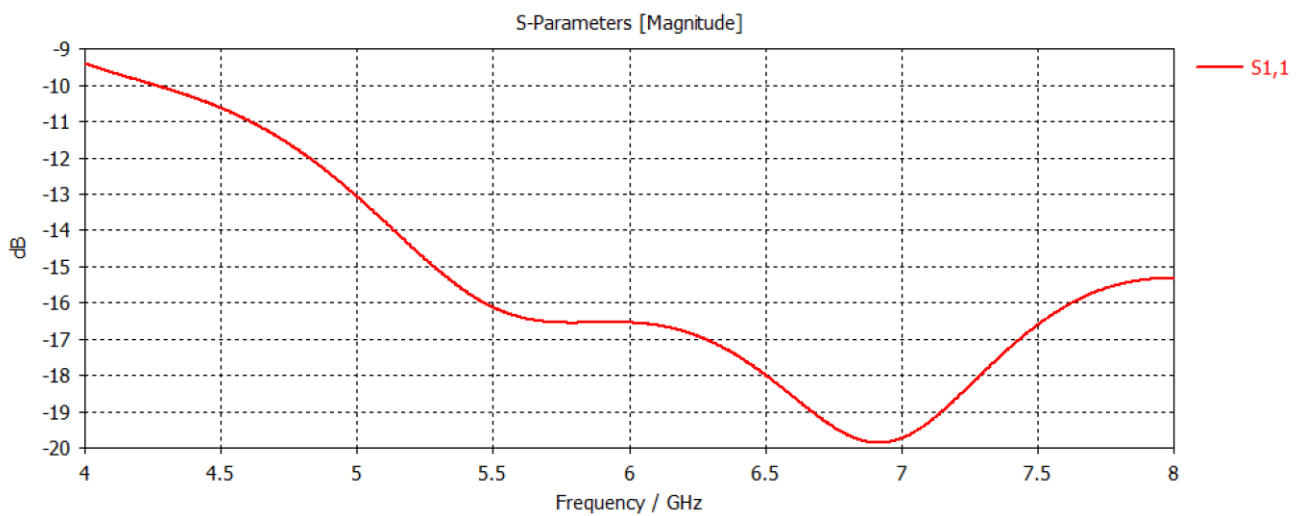




S_{11} for 100 ohm

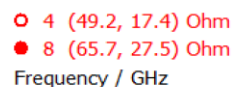


S_{11} for 0.52pF

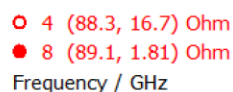


S_{11} for 33 ohm and 0.75pF

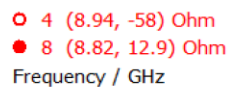
(i) Smith Chart



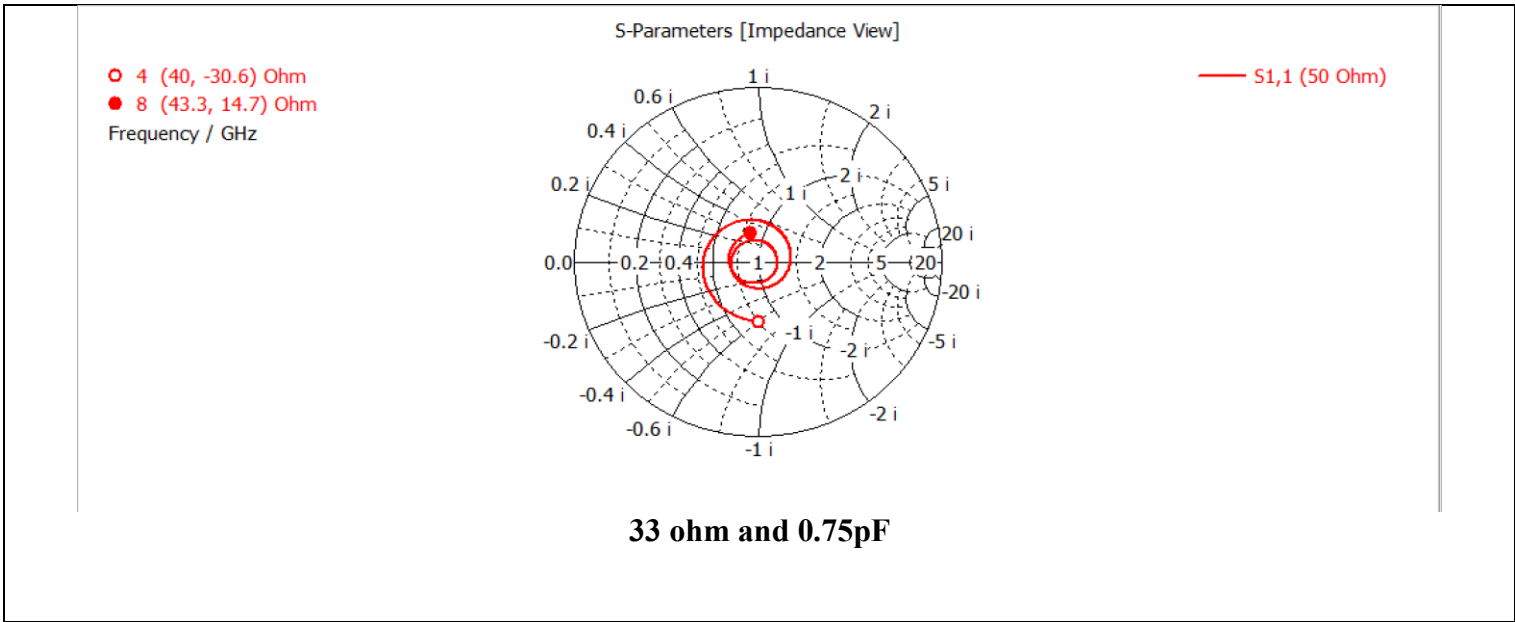
50 ohm



100 ohm



0.52 pF



Conclusions:

In the first case, where the load is matched, the S_{11} is consistently lower than -10dB (i.e. the power reflected back is less than 10% of input power) for all frequencies. On the smith chart, you can observe that the circle traced out by the normalized impedance is relatively close to the center and is also the circle with the smallest radius compared to the other cases. This shows that when you have a matched load, the reflectance is minimized with large amount of power transmitted successfully.

Increasing the resistance to 100 ohm, ($\tilde{Z}_L = 2$), we observe the S_{11} values become less negative, with the worst values lying at about -9.6dB. The smith chart also shows an expanded circle traced out, indicating that the reflectance has increased. Hence, we conclude that for a mismatched case, we have more power being reflected back and hence less power transferred overall.

Switching to a purely capacitive load, such that the reactance is about 50 ohms at 6 GHz, we observe much worse S_{11} graphs, indicating that a large amount of power is being reflected away. Here the S_{11} values are as high as -1dB. This worse performance is also reflected in the smith chart, with it tracing a much wider circle now.

Switching to a mix of real and capacitive load, choosing values such that the magnitude of the impedance is equal to Z_o , i.e by setting $R = \frac{50}{\sqrt{2}}$ and $X_L = \frac{50}{\sqrt{2}}$ (at 6 GHz), we see that the modulus of the reflectance is reduced than what it were in the case of a purely capacitive load even though $|Z_L|$ is the same in both cases.

Load Case	S11 Range (dB)	Reflected Power Range (%)	Transmitted Power Range (%)
Matched Load (50Ω)	-11 to -16 dB	2.51% – 7.94%	92.06% – 97.49%
Mismatched (100Ω)	-9.7 to -11 dB	7.94% – 10.72%	89.28% – 92.06%
Purely Capacitive	-1.3 to -2.9 dB	51.29% – 74.13%	25.87% – 48.71%
Mixed (R+C)	-9.1 to -20 dB	1.00% – 12.30%	87.70% – 99.00%

EXPERIMENT NO. 4

Implementation of a Quarter-wavelength line

Date: 01/07/2025

Group No. 2

Name: Mihika Mundhada

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Name: Pramit Pal

ID No: 2023AAPS0765P

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ID No: 2023AAPS0772P

Name: Nikhil Matthew

ID No: 2023AAPS0730P

Aim: Design a quarter-wavelength line and demonstrate matching for real-to-real loads

- To find the corresponding length and width of the quarter wavelength section at a frequency of operation, and plot the scattering parameter (S_{11})

Equipment & Components: CST Studio

Theory:

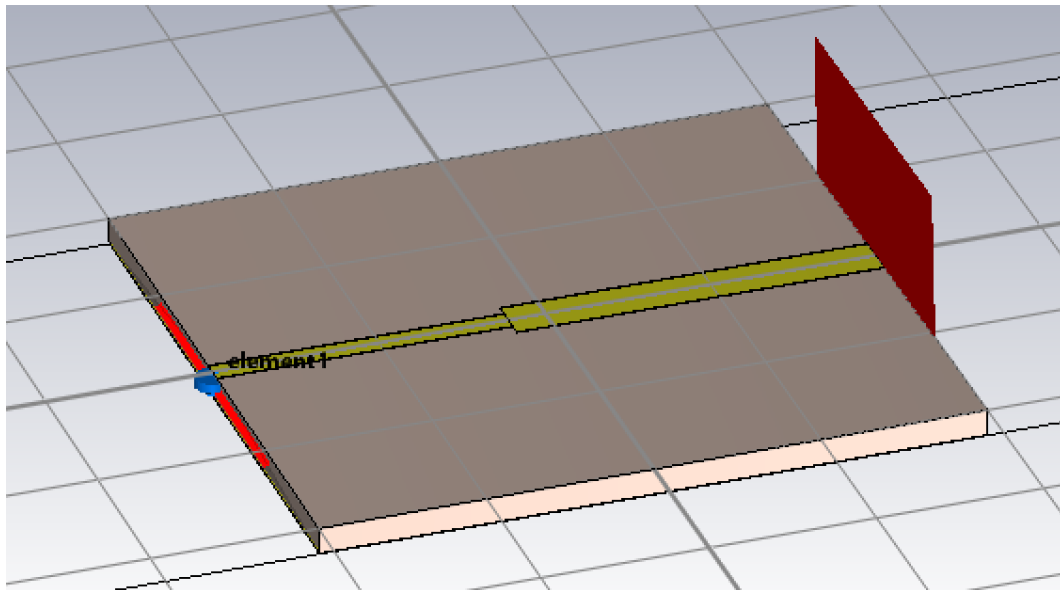


Figure 1 – Quarter wave antenna as setup in CST with one waveguide port and one lumped element

A quarter wave antenna is a basic form of impedance matching, constructed by inserting a transmission line of length equal to one quarter of the guided wavelength between the feed and load. The dielectric constant of the medium and the operating frequency dictates the length needed. The physical length of the quarter wave section is determined with the following relations

$$\lambda = \frac{c}{f\sqrt{\epsilon_r}}, \quad l = \frac{\lambda}{4}$$

The width of the quarter wave length antenna is determined by the load impedance the previous transmission line impedance. At the beginning of the quarter wave section the, impedance as seen by the transmission line Z_{in} must match the impedance of the transmission line Z_o . To do so we calculate the impedance of the quarter wave section, Z_{qw} . For a quarter wave section:

$$Z_o = Z_{in} = \frac{Z_{qw}^2}{Z_L}$$

$$Z_{qw} = \sqrt{Z_o * Z_L}$$

With the length and the impedance of the quarter wave section given, we achieve load matching.

Design Problem:

For 2 GHz operation, with transmission line impedance of 50 ohm and load impedance of 100 ohm, we calculate

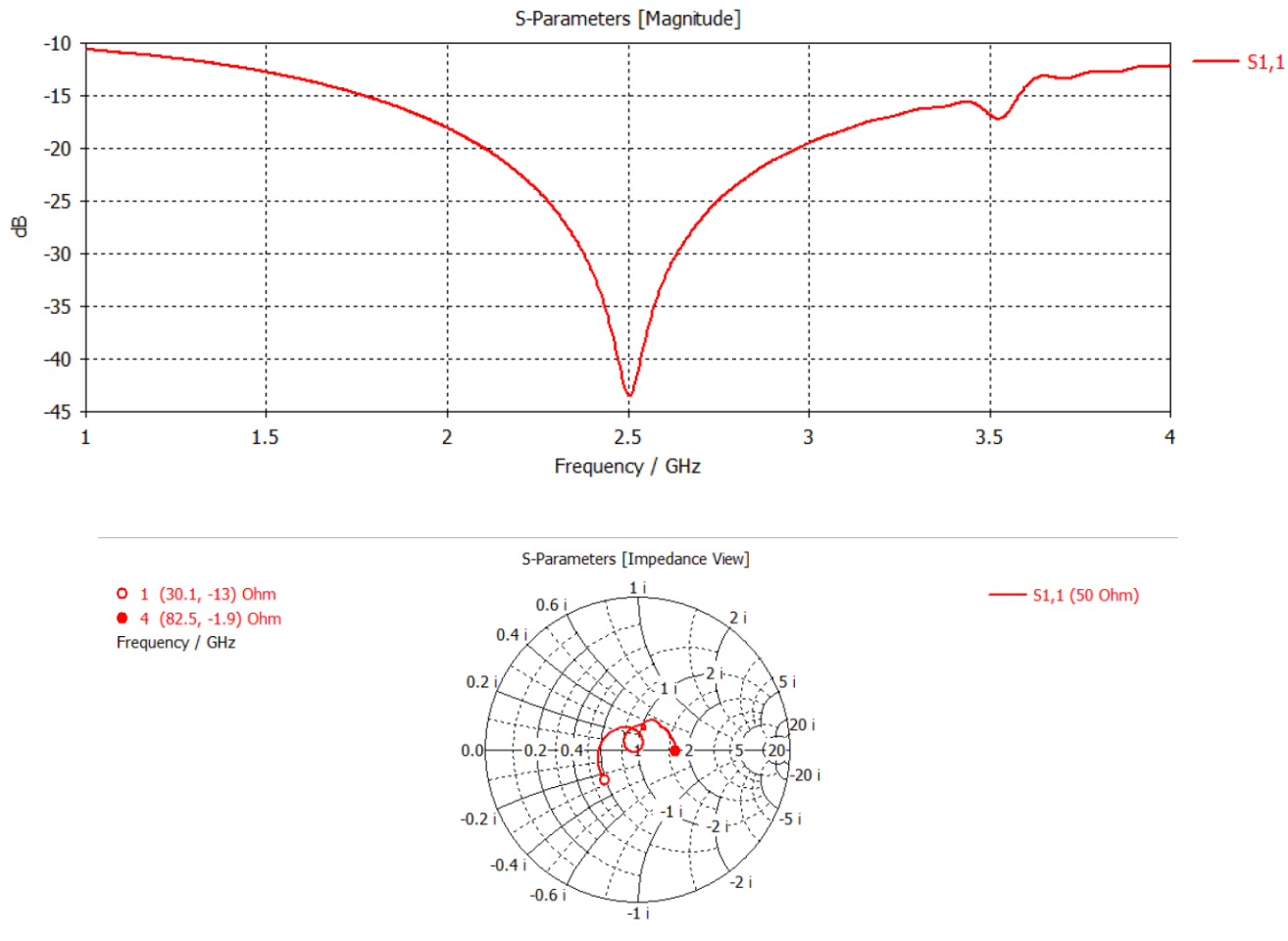
$$l = 17.77 \text{ mm}$$

$$Z_{qw} = 70.7 \text{ ohm}$$

$$w = 0.82 \text{ mm}$$

Results:

S_{11} parameters



Conclusions:

From the S_{11} parameter graphs we can conclude that at the target frequency we achieve high matching with the value being around -43 dB, indicating that only **0.005%** of the power is reflected with the rest **99.995%** being transmitted. We can also observe that this low reflectance has a peak at around our target frequency, with the quality of transmission dropping at nearby frequencies. This arises from the fact that the length of the quarter wave section is no longer the quarter wave for other frequency values. The Smith Chart also shows similar results.

EXPERIMENT NO. 1

STUDY OF MICROWAVE COMPONENTS & EQUIPMENT

Date: 19/8/2025	Group No. 2
Name: Saikrishna Bulusu	ID No: 2023AAPS0728P
Name: Mihika Mundhada	ID No: 2023AAPS0752P
Name: Somesh Huljute	ID No: 2023AAPS0772P
Name: Sarthak Sur	ID No: 2023AAPS0731P
Name: Pramit Pal	ID No: 2023AAPS0765P

Aim: To become familiar with microwave components and instruments available in the laboratory.

Equipment & Components: Klystron power supply, Gunn power supply, Frequency meter, VSWR meter, Klystron Tube, Gunn diode, PIN modulator, Isolator, E plane Tee, H plane Tee, Directional coupler, Matched termination, movable short, horn antenna etc.

PROCEDURE:

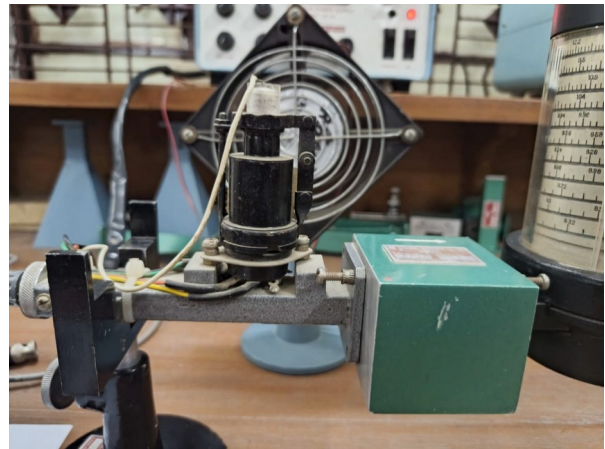
- Note relevant technical specifications of the instruments.
- Study position and functions of the front panel controls of the equipment.
- Know basic principle of operation and functional block diagram of the instrument.
- Facilities provided and limitations of the equipment if any.
- Know initial settings of controls of the equipment before switching on the supply.
- Precautions to be taken while carrying out the measurements.

Observations:

1. Identification through labels is provided on each component.



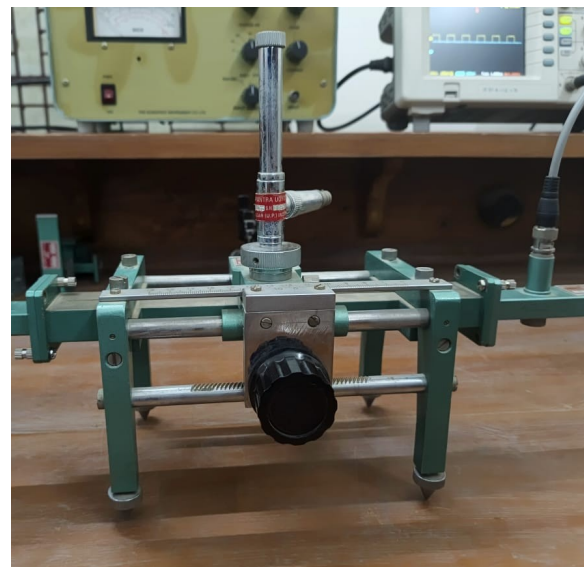
Klystron Power Supply



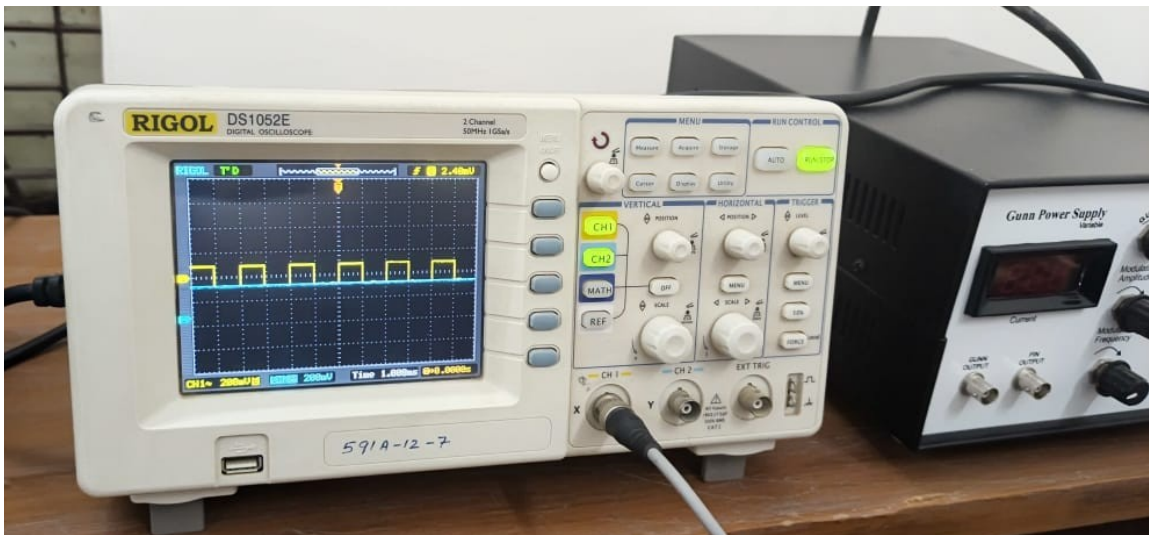
Klystron (left) and isolator(right)



Frequency Meter



Slotted Section and Tunable Probe



Digital Signal Oscilloscope (left) and Gunn Power Supply

2. Initial conditions and precautions:

- Do not touch Klystron tube as it may give shock if black wired cap is misplaced.
- Bench setup should be in proper order like all stickers must be on front side.
- Don't try to see the signal inside guide as it may be absorbed by body and may cause some side effects.
- Set micrometer of variable attenuator at 11 mm approx. As height of rectangular waveguide is 10.02mm in this position of micrometer it offers less attenuation.
- Never keep Frequency meter needles closed to lock.
- If we don't require tunable probe remove and cover with cap to protect needle otherwise keep as it is in setup and adjust at proper position.
- Before switching ON klystron power supply make sure Beam voltage knob at min position, Reflector voltage knob at max position.
- Before switching ON VSWR meter make sure gain control knobs at mid position, input switch low impedance and lower position Range selector.
- Before switching OFF, make sure all knob positions remain at their minimum levels.

3. Function of Each Equipment/Component:

- **Klystron Tube:** A high-power microwave amplifier that uses electron beam velocity modulation to generate or amplify microwave signals. It serves as a stable, high-frequency source in microwave systems and can produce significant power output for various applications.
- **Klystron Mount:** A mechanical housing that securely holds the klystron tube and provides proper electrical connections to the waveguide system. It ensures optimal coupling between the klystron and the transmission line while maintaining safety and stability.
- **Gunn Oscillator:** A solid-state microwave source that generates continuous wave signals using the Gunn effect in semiconductor materials like gallium arsenide. It provides a compact, reliable alternative to vacuum tube oscillators for lower power applications.
- **PIN Modulator:** Uses PIN diodes to control microwave signal amplitude or provide on/off switching. The modulator allows for precise control of signal power and can create pulsed microwave signals from continuous wave sources.
- **VSWR Meter:** Measures the Voltage Standing Wave Ratio in transmission lines, indicating how well the load is matched to the source. It helps identify impedance mismatches and reflections in the microwave system.
- **Isolator:** A non-reciprocal device that allows microwave energy to pass in only one direction while blocking reflections from reaching the source. This protects sensitive components like oscillators from damage due to reflected power.
- **Variable Attenuator:** Provides adjustable attenuation of microwave signals, allowing precise control of power levels throughout the system. It's essential for calibration and ensuring proper signal levels for measurements.
- **Frequency Meter:** A resonant cavity device used to accurately measure the frequency of microwave signals. It typically uses a calibrated cavity with a tuning mechanism to determine operating frequencies.
- **Slotted Section:** A section of waveguide with a longitudinal slot that allows a probe to sample the electric field. It's used to measure standing wave patterns and determine impedance characteristics of loads and components.
- **Tunable Probe:** A movable probe that can be inserted into the slotted section to detect electric field intensity at different positions. It enables measurement of standing wave patterns and voltage distribution along the transmission line.
- **Detector Mount:** Houses a crystal detector diode that converts microwave energy into a DC signal proportional to the microwave power. This allows measurement and monitoring of microwave power levels using conventional meters.
- **Matched Termination:** Provides a load with characteristic impedance matching the transmission line, absorbing all incident power without reflections. It's used as a reference load and to terminate unused ports.
- **E-plane TEE:** A three-port junction where the branching arm extends in the direction of the electric field. It acts as a power divider/combiner and is commonly used in microwave circuits for signal splitting.
- **H-plane TEE:** A three-port junction where the branching arm extends perpendicular to the electric field direction. It provides different phase relationships compared to E-plane TEE and is used for specific circuit applications.
- **Magic TEE:** Combines both E-plane and H-plane TEE characteristics in a four-port junction. It provides isolation between certain ports and is widely used in balanced mixers, bridges, and antenna feed systems.
- **Fixed Attenuator:** Provides a predetermined, constant level of signal attenuation. It's used for power level adjustment, impedance matching, and protecting sensitive components from excessive power levels.
- **Wave Guide:** A hollow metallic conduit that guides electromagnetic waves from one point to another with minimal loss. It serves as the primary transmission medium for microwave energy in the system.

- **Solid Dielectric Cell:** Contains dielectric material samples for measuring their electrical properties at microwave frequencies. It allows determination of dielectric constant and loss tangent of various materials.
- **M.H.D Coupler (Multi-Hole Directional Coupler):** Samples a portion of the forward or reverse traveling wave without significantly affecting the main signal. It's used for power monitoring and signal injection in microwave systems.
- **T-circulator:** A three-port non-reciprocal device that routes microwave energy in a circular pattern from port to port. It's commonly used in radar systems and provides isolation between transmitter and receiver.
- **Y-circulator:** Similar to T-circulator but with a Y-shaped configuration. It provides the same non-reciprocal routing function but with different physical geometry for specific applications.
- **Phase Shifter:** Introduces a controlled phase delay to microwave signals without significantly affecting amplitude. It's essential for phased array antennas and various microwave measurement applications.
- **Standard Gain Antenna:** A reference antenna with precisely known gain characteristics used for calibration and measurement purposes. It provides a standard for comparing other antennas and measuring field strength.
- **Sectorial Horn – H-plane:** A horn antenna that flares only in the H-plane direction, providing directional radiation with specific beamwidth characteristics. It's used for moderate gain applications and as a feed for larger antennas.
- **Sectorial Horn – E-plane:** Similar to H-plane horn but flares only in the E-plane direction, creating a different radiation pattern. It provides complementary characteristics to H-plane horns for various applications.
- **Pick up Horn:** A receiving antenna designed to intercept and collect microwave energy for measurement purposes. It converts electromagnetic waves back into guided waves for detection and analysis in the measurement system.

Conclusions:

Block Diagram

